

Unit:8

Distributed Operating System

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**MCA FIRST SEMESTER
OPERATING SYSTEM
MCA 103**

Operating System

Concept of Distributed Operating System

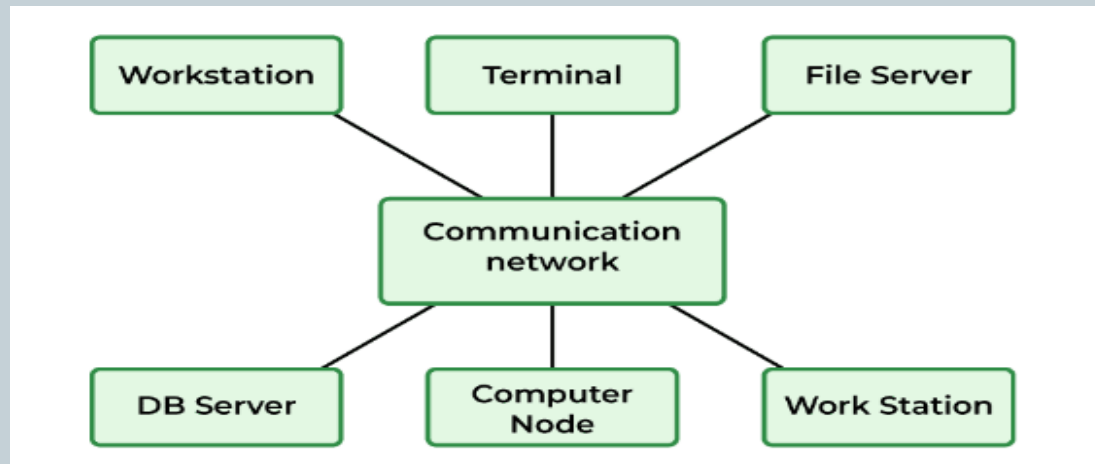
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- A distributed operating system is one in which several computer systems connected through a single communication channel.
- Distributed OS runs on multiple CPUs but for an end-user, it is just an ordinary centralized operating system.
- It can share all resources like CPU, disk, network interface, nodes, computers, etc. from one site to another site, and it increases the data available on the entire system.
- Effective communication channels like high-speed buses and telephone lines connect all processors, each equipped with its own local memory and other neighboring processors. Due to its characteristics, a distributed operating system is classified as a loosely coupled system

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Concept of Distributed Operating System

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- The above figure illustrates the structure of a distributed operating system.

Real Life Example of Distributed OS: Banking system

- Suppose there is a bank whose headquarters is in Kathmandu. That bank has branch offices in cities like Lumbini, Pokhara, Damak, and Surkhet. You can operate your bank by going to any of these branches. How is this possible? It's because whatever changes you make at one branch office are reflected at all branches. This is because of the distributed system.

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Advantages of Distributed OS

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- ❖ **High Availability** – Distributed operating systems are designed to keep applications running even if a single computer fails. This means that if one computer goes down, the other computers in the system can take over, ensuring that your applications continue to function.
- ❖ **Scalability** – Distributed operating systems can be easily scaled up or down depending on our needs. If we need more computing power, we can add more computers to the system. If we need less, we can remove them. This makes it easier to manage your resources and keep costs down.
- ❖ **Resource sharing** – With a distributed operating system, resources such as memory, processing power, and storage can be shared among different computers in the system. This means that we can use the resources of other computers in the network when we need them, which can increase efficiency and reduce costs.

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Advantages of Distributed OS

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- ❖ **Improved performance** – By distributing tasks across multiple computers, a distributed operating system can improve the performance of your applications. This is because each computer in the network can focus on a specific task, rather than having to share resources with other tasks on the same machine.
- ❖ **Fault tolerance** – Distributed operating systems are designed to be fault tolerant, which means they can continue to function even if some of the computers in the system fail. This is because the system is designed to distribute tasks across multiple computers, so if one computer fails, the other computers can take over.

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Dis-advantages of Distributed OS

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- ❖ **Increased Complexity-** Distributed Operating System involves managing multiple computers that communicate and coordinate with each other. This makes it a very complex system, which is difficult to manage and troubleshoot.
- ❖ **Security Concerns** – With the number of computers and users that are part of a Distributed Operating System, there are many security risks involved. If one computer is compromised, it can affect the entire system, making it vulnerable to cyber-attacks and data breaches.
- ❖ **Network Dependency** – The proper functioning of a Distributed Operating System depends on the network. If the network goes down or experiences latency issues, it can affect the performance of the entire system.
- ❖ **High Cost** – Distributed Operating System requires a lot of resources, including hardware, software, and skilled personnel. This can make it expensive to implement and maintain.
- ❖ **Compatibility Issues** – Not all applications are designed to work with Distributed Operating Systems. This can create compatibility issues and make it difficult to use certain applications in a distributed environment.

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Types of Distributed OS

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There are mainly two types, they are as follows:

1. Client/Server Systems

- In this system, the client requests the server for a resource.
- On the other hand, the server provides this resource to the client.
- One client contacts only a single server at a time.
- Whereas a single server can deal with multiple clients simultaneously.
- The clients and servers connect through a computer network in the system.

2. Peer to Peer Systems

- In this system, the nodes play an important role.
- All the work equally divides among the nodes.
- Furthermore, these nodes can share data or resources as per the requirement. Again, they require a network to connect.

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Design Issues in Distributed OS

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The main design issues in distributed OS are as follows:

1. Concurrency Control:

Managing simultaneous access to shared resources without conflicts or data corruption.

2. Fault Tolerance:

Ensuring system stability and functionality even when some components fail or behave unexpectedly.

3. Communication Protocols:

Designing efficient, reliable, and secure communication channels among distributed entities.

4. Synchronization:

Coordinating actions across distributed components to maintain consistency and avoid race conditions.

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Design Issues in Distributed OS

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5. Security:

Implementing measures to protect data, prevent unauthorized access, and ensure secure communication.

6. Scalability:

Designing the system to handle growing numbers of users, resources, and transactions efficiently.

7. Resource Management:

Efficient allocation and de-allocation of resources such as CPU, memory, and storage among distributed nodes.

8. Naming and Addressing:

Consistently identifying and locating resources in a distributed environment.

9. Consistency Models:

Defining rules for maintaining data consistency across distributed nodes, especially in the presence of failures.

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Communication in Distributed OS

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- In distributed system, processes running on separate computers cannot directly access each other's memory.
- There are two approaches to communication in distributed OS.

1. Shared Memory

- Process must have access to some form of shared memory(i.e. they must be threads, they must be process that can share memory , or they must have access to shared resources, such as file).
- In distributed shared memory, the processes on separate computers all have access to the same virtual address space. The memory pages that make up this address space actually reside on separate computers.

2. Message Passing

- In message passing, processes communicate by sending messages to each other.

Communication in Distributed OS

Interprocess communication (IPC)

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Introduction

- **IPC** is at the heart of all distributed systems, so we need to know the ways that processes can exchange information.
- Communication in distributed systems is based on **Low-level message passing** as offered by the underlying network.

Layered Protocols

Two processes **A,B**

A wants to communicate with **B**.

A builds a message in its own address space.

A execute a call to the OS to send the message.

To prevent chaos (mass)

A,B have to agree on the meaning of the bites being sent.

Communication in Distributed OS

Interprocess communication (IPC)

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- The International Standards Organization (ISO) developed a **reference model** that clearly identifies the various levels involved **Open Systems Interconnection (OSI)**.
- It provides a description of how network hardware and software work together in a layered fashion to make communications possible.

Communication in Distributed OS

Interprocess communication (IPC)

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- Communication divided into **7 layers**.
- Each layer has a responsibility to perform specific tasks but all of the layers are needed for a message to reach its destination.
- Message **send** (downwards) Message **received** (upwards).

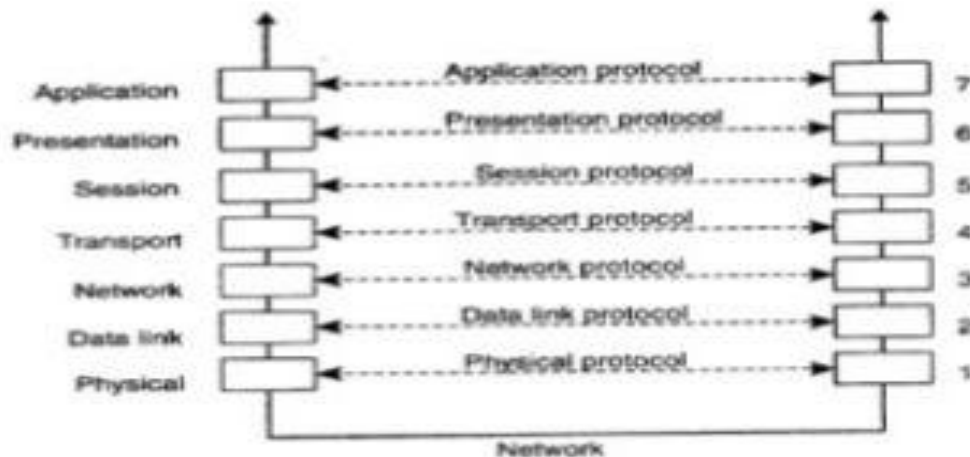


Figure 4-1. Layers, interfaces, and protocols in the OSI model.

Communication in Distributed OS

Interprocess communication (IPC)

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Two general type of protocols

- Connection oriented protocols:

Before exchanging data the sender and receiver first establish a connection, and possibly negotiate the protocol they will use When they are done, they must terminate the connection.

The Telephone

- Connectionless protocols:

No setup in advance is needed, The sender just transmits the first message when it is ready.

The mailbox or mail

Communication in Distributed OS

Interprocess communication (IPC)

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Layered Protocols

Application Layer

Presentation Layer

Session Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer

Communication in Distributed OS

Interprocess communication (IPC)

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Application Layer

- It is the software that the end-user interacts with.
- It serves as a window through which application processes can access network services.



Presentation Layer

- Think of it as the network's translator.
- It is responsible for translating the data, encrypting the data, changing or converting the character set.
- Manages data compression to reduce the number of bits that need to be transmitted.

Communication in Distributed OS

Interprocess communication (IPC)

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Session Layer

- Provides dialog control between communicating processes, such as which party is currently talking, when, and for how long.
- Allows users to insert checkpoints into long transfers.
- In the event of crash, it is necessary to go back only to the last checkpoint.

Transport Layer

- Turns the underlying network into something that an application developer can use.
- Ensures that packets are delivered error free, in sequence, and without losses or duplications.

Communication in Distributed OS

Interprocess communication (IPC)

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Network Layer

- Addresses the packets and translating logical addresses and names into **physical addresses**.
- Determines the route from the source to the destination computer (**Delay-Wise**).

Data Link Layer

- Groups bits into frames and sees that each frame is correctly received (checksum), If checksums differ, requests a retransmission.
- Sends data frames from the network layer to the physical layer



Communication in Distributed OS

Interprocess communication (IPC)

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Physical Layer

- The physical layer is totally **hardware-oriented**.
- It is responsible for transmitting bits (zeros and ones) from one computer to another.
- Defines the duration of each impulse.

Synchronization in Distributed OS

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- A synchronization in distributed systems is a process of sharing information between multiple computers so that they can work together.
- This is often done by sharing a common clock so that all computers can keep track of time.
- Synchronization can also be used to share data between computers.
- In the distributed system, the hardware and software components communicate and coordinate their actions by message passing.
- Each node in distributed systems can share its resources with other nodes. So, there is a need for proper allocation of resources to preserve the state of resources and help coordinate between the several processes.
- To resolve such conflicts, synchronization is used.
- Synchronization in distributed systems is achieved via clocks.
- The physical clocks are used to adjust the time of nodes. Each node in the system can share its local time with other nodes in the system.

Synchronization in Distributed OS

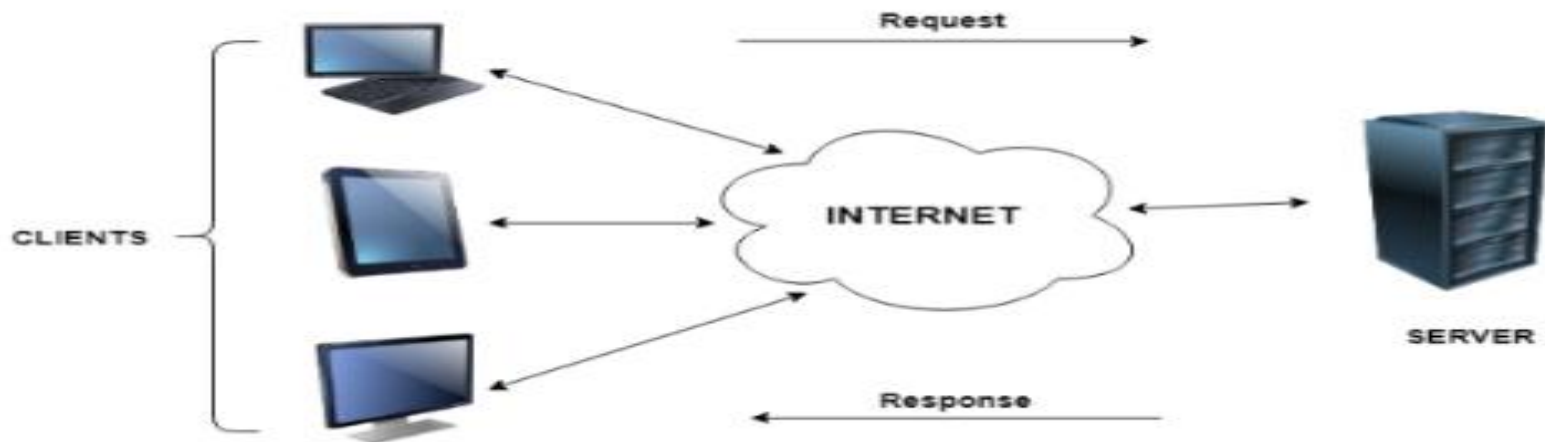
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- The time is set based on UTC (Universal Time Coordination). UTC is used as a reference time clock for the nodes in the system.
- Clock synchronization can be achieved by 2 ways: External and Internal Clock Synchronization.
- ❖ **External clock synchronization** is the one in which an external reference clock is present. It is used as a reference and the nodes in the system can set and adjust their time accordingly.
- ❖ **Internal clock synchronization** is the one in which each node shares its time with other nodes and all the nodes set and adjust their times accordingly.

Client Server Computing

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- Client-server computing is a method of distributed computing where one program (a client) makes a request to be fulfilled by another program (a server).
- A server may serve multiple clients at the same time while a client is in contact with only one server.
- Both the client and server usually communicate via a computer network (in a distributed system) but sometimes they may reside in the same system.
- The server is more powerful than the client since the server has to respond the request provided by all the clients.



Advantages and Dis-advantages of Client Server Computing

■ Advantages include

- Centralization - Resources and data security are controlled through the server.
- Scalability - Any or all elements can be replaced individually as needs increase.
- Flexibility - New technology can be easily integrated into system.
- Interoperability - All components (client/network/server) work together.
- Accessibility - Server can be accessed remotely and across multiple platforms.

■ Disadvantages include

- Expense - Requires initial investment in dedicated server.
- Maintenance - Large networks will require a staff to ensure efficient operation.
- Dependence - When server goes down, operations will cease across the network.

Mutual Exclusion algorithm for Distributed OS

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- When a process wants to enter a critical region, it builds a message containing the name of the critical region it wants to enter, its process number, and the current time.
- It then sends the message to all other processes, conceptually including itself.
- The sending of messages is assumed to be reliable; that is, every message is acknowledged.
- Reliable group communication if available, can be used instead of individual messages.
- When a process receives a request message from another process, the action it takes depends on its state with respect to the critical region named in the message.
- Three cases have to be distinguished:
 - If the receiver is not in the critical region and does not want to enter it, it sends back an OK message to the sender.
 - If the receiver is already in the critical region, it does not reply. Instead, it queues the request.
 - If the receiver wants to enter the critical region but has not yet done so, it compares the timestamp in the incoming message with the one contained in the message that it has sent everyone.
 - The lowest one wins. If the incoming message is lower, the receiver sends back an OK message.
 - If its own message has a lower timestamp, the receiver queues the incoming request and sends nothing.

Mutual Exclusion algorithm for Distributed OS

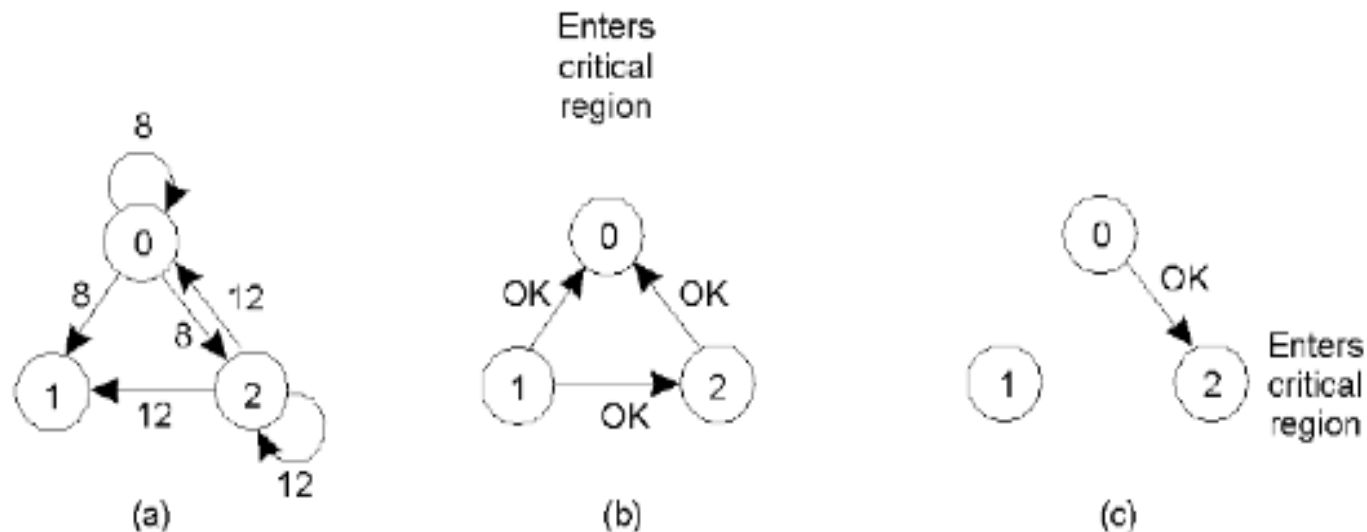


Figure (a) Two processes want to enter the same critical region at the same moment.
(b) Process 0 has the lowest timestamp, so it wins. (c) When process 0 is done, it sends an OK also, so 2 can now enter the critical region.

- Process 0 sends everyone a request with timestamp 8.
- while at the same time, process 2 sends everyone a request with timestamp 12.
- Process 1 is not interested in entering the critical region, so it sends OK to both senders.
- Processes 0 and 2 both see the conflict and compare timestamps.
- Process 2 sees that it has lost, so it grants permission to 0 by sending OK.
- Process 0 now queues the request from 2 for later processing and enters the critical region.
- When it is finished, it removes the request from 2 from its queue and sends an OK message to process 2, allowing the latter to enter its critical region.

Deadlock in Distributed OS

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Deadlock

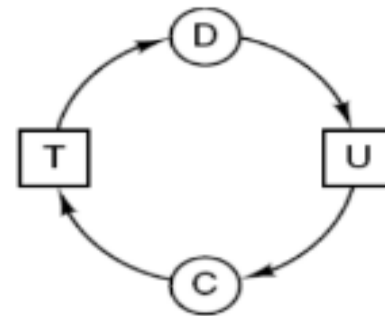
- A set of processes is deadlocked if each process in the set is waiting for an event that only another process in the set can cause.
- Suppose process D holds resource T and process C holds resource U.
- Now process D requests resource U and process C requests resource T but none of process will get the resource because these resources are already hold by other process so both can be blocked, with neither one be able to proceed, this situation is called deadlock.
- Example:



(a)



(b)



(c)

Figure: Resource allocation graphs. (a) Holding a resource. (b) Requesting a resource. (c) Deadlock.

- As shown in above figure, resource T assigned to process D and resource U is assigned to process C.
- Process D is requesting / waiting for resource U and process C is requesting / waiting for resource T. Processes C and D are in deadlock over resources T and U.

Deadlock in Distributed OS

Conditions that lead to deadlock

- There are four conditions that must hold for deadlock:
 - 1) Mutual exclusion condition
 - Each resource is either currently assigned to exactly one process or is available.
 - 2) Hold and wait condition
 - Process currently holding resources granted earlier can request more resources.
 - 3) No preemption condition
 - Previously granted resources cannot be forcibly taken away from process.
 - 4) Circular wait condition
 - There must be a circular chain of 2 or more processes. Each process is waiting for resource that is held by next member of the chain.
 - All the four of these conditions must be present for a deadlock to occur.
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Distributed File System (DFS)

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- A Distributed File System (DFS) is a file system that is distributed on multiple file servers or multiple locations.
- Files are accessible just as if they were stored locally, from any device and from anywhere on the network.
- A DFS makes it convenient to share information and files among users on a network in a controlled and authorized way.
- A DFS is also called a client-server architecture based application, which allows the user or clients to access the data from the server as it is stored in their own computer.
- It provides location transparency and redundancy help to improve the data availability.
- It also use data replication strategy on multiple servers to prevent data access failure.

Benefits of Distributed File System (DFS)

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- ❖ **Flexibility in storage management** – In DFS, storage management is very flexible and we can easily modify it according to our need.
- ❖ **Load sharing advantage** – Load sharing can be done with optimal results using the DFS. Load sharing is one of the best benefits of DFS.
- ❖ **Security Integration** – If we want to implement security then it can be easily done in the DFS.
- ❖ **Graphical way of Administration** – Graphical view of administration window is available here, which reduces cost in administration training.
- ❖ **High Availability** – High availability is also one of the best benefits of DFS. It keeps all the important data available all the time.

Features of Distributed File System (DFS)

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The features of DFS are as follows –

- ❖ **User mobility**
- ❖ **Easy to use**
- ❖ **High availability**
- ❖ **Performance**
- ❖ **Coherent access**
- ❖ **Location independence**
- ❖ **File locking**
- ❖ **Multi-networking access**
- ❖ **Local gateways**
- ❖ **Multi-protocol access**

Security in Distributed OS

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- Security in a distributed system poses unique challenges that need to be considered when designing and implementing system.
- Security in distributed systems introduces two specific concerns that centralized systems do not have.
 - ❖ The first is the use of a network where contents may be seen by other, possibly malicious, parties.
 - ❖ The second is the use of servers. Because clients interact with services (applications) running on a server, the application rather than the operating system is responsible for authenticating the client and controlling access to services.
- In a distributed system, one must consider many possible security risks. To mitigate these risks there are a number of strategies that can be employed:
 - ❖ Encryption algorithms that protect data in transit and at rest.

Security in Distributed OS

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- ❖ Firewalls that limit access to specific ports/cables.
- ❖ Intrusion detection systems that identify anomalous behavior among network services.
- ❖ Intrusion prevention systems (IPS) respond to attempted intrusions by initiating defensive actions like blocking suspicious IP addresses or taking down compromised servers.

END OF UNIT 8